A One-Dimensional Counterexample to $G_M = F_M$ for CMQd Sets

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Status. Candidate full counterexample, likely valid, for expert review.

1 Source question

Let E be a real Banach space, $B = E \times E^*$, and

$$q(x, x^*) = \langle x, x^* \rangle, \qquad r = q + \tfrac{1}{2}\| \cdot \|^2.$$

The source calls a subset $M \subset B$ *CMQd* when it is closed, monotone, and quasidense, the last condition meaning

$$\inf_{m \in M} r(m - b) = 0 \qquad (b \in B).$$

Writing $L(x, x^*) = (x^*, \widehat{x})$ and $\widetilde{q}(y^*, y^{**}) = \langle y^*, y^{**} \rangle$, it defines, for a maximally monotone M,

$$\begin{aligned}
P_M(b) &= -\inf_{m \in M} q(m - b), \\
G_M(b^*) &= -\inf_{m \in M} \widetilde{q}(Lm - b^*), \\
F_M(b^*) &= -\inf_{b \in B} \big[P_M(b) + \widetilde{q}(b^* - Lb) \big].
\end{aligned}$$

Every CMQd set is maximally monotone [1, Theorem 4.2]. The source proves $G_M \le F_M$ and asks in Remark 11.4 whether equality holds for every CMQd set [1, Remark 11.4, PDF p. 20].

> **Remark 11.4.** A number of the results that we have proved lead to the conjecture that *if M a CMQd subset of B then $G_M = F_M$ on B^*.* We do not know if this is true.

2 Claimed contribution

Theorem 1. *Let $E = \mathbb{R}$ with its usual norm. For every $\lambda > 0$, the line*

$$M_\lambda = \{(s, \lambda s) : s \in \mathbb{R}\} \subset B = \mathbb{R}^2$$

is CMQd. Under the canonical identifications $B^* = \mathbb{R}^2$ and $L(u,v) = (v,u)$, its three source-paper functions are

$$P_{M_\lambda}(u,v) = \frac{(v-\lambda u)^2}{4\lambda}, \tag{1}$$

$$G_{M_\lambda}(y,z) = \frac{(y-\lambda z)^2}{4\lambda}, \tag{2}$$

$$F_{M_\lambda}(y,z) = \begin{cases} 0, & y = \lambda z, \\ +\infty, & y \neq \lambda z. \end{cases} \tag{3}$$

Consequently $G_{M_\lambda} \neq F_{M_\lambda}$. In particular, for $\lambda = 1$ and $b^* = (0,1)$,

$$G_{M_1}(0,1) = \frac{1}{4} \qquad \textit{but} \qquad F_{M_1}(0,1) = +\infty.$$

Thus the conjecture in Remark 11.4 is false, already in dimension one.

3 Proof intuition

The penalty $P_{M_\lambda}(u,v)$ controls only the transverse coordinate $v - \lambda u$. In the definition of F_M, the remaining quadratic expression instead contains $(\lambda u + v)^2$. It therefore has a null direction $(u,v) = (t, -\lambda t)$. Unless the dual point lies on the line $y = \lambda z$, the linear part drifts to $-\infty$ along this direction, forcing $F_M = +\infty$. Direct minimization over the graph defining G_M has no such escape and always gives a finite quadratic value.

4 Proof of Theorem 1

Fix $\lambda > 0$. The set M_λ is closed. It is monotone because

$$q\big((s,\lambda s) - (t,\lambda t)\big) = \lambda(s-t)^2 \geq 0.$$

For an arbitrary $b = (a,c) \in B$,

$$\begin{aligned} r\big((s,\lambda s) - (a,c)\big) &= (s-a)(\lambda s - c) + \frac{1}{2}\big[(s-a)^2 + (\lambda s - c)^2\big] \\ &= \frac{1}{2}\big[(1+\lambda)s - (a+c)\big]^2. \end{aligned}$$

Its infimum over $s \in \mathbb{R}$ is zero, attained at $s = (a+c)/(1+\lambda)$. Hence M_λ is quasidense and therefore CMQd.

For $(u,v) \in B$, completing the square gives

$$(s-u)(\lambda s - v) = \lambda\left(s - \frac{v+\lambda u}{2\lambda}\right)^2 - \frac{(v-\lambda u)^2}{4\lambda}.$$

Taking the negative infimum proves (1).

Now let $b^* = (y,z) \in B^*$. Since $L(s,\lambda s) = (\lambda s, s)$,

$$(\lambda s - y)(s - z) = \lambda\left(s - \frac{y+\lambda z}{2\lambda}\right)^2 - \frac{(y-\lambda z)^2}{4\lambda}.$$

The definition of G_M immediately gives (2).

It remains to compute F_M. For $b = (u, v)$, the expression whose infimum occurs in its definition is

$$\begin{aligned} J(u, v) &:= P_{M_\lambda}(u, v) + \widetilde{q}\big((y, z) - L(u, v)\big) \\ &= \frac{(v - \lambda u)^2}{4\lambda} + (y - v)(z - u) \\ &= \frac{(\lambda u + v)^2}{4\lambda} - yu - zv + yz. \end{aligned} \tag{4}$$

Along $(u, v) = (t, -\lambda t)$ this becomes

$$J(t, -\lambda t) = (\lambda z - y)t + yz.$$

If $y \neq \lambda z$, this affine function is unbounded below, so $\inf J = -\infty$ and $F_{M_\lambda}(y, z) = +\infty$.

If $y = \lambda z$, put $w = \lambda u + v$ in (4). Then

$$J(u, v) = \frac{w^2}{4\lambda} - zw + \lambda z^2 = \frac{(w - 2\lambda z)^2}{4\lambda}.$$

Its infimum is zero. Thus $F_{M_\lambda}(y, z) = 0$, proving (3) and the theorem.

5 Stress checks and scope

The value $+\infty$ in (3) is permitted by the source's stated codomain $F_M : B^* \to]-\infty, \infty]$. No nonreflexive phenomenon, closure subtlety, or unattained infimum is involved. For the displayed witness, one can check the divergence without any parameter algebra:

$$P_{M_1}(u, v) + (0 - v)(1 - u) = \frac{(u + v)^2}{4} - v.$$

At $(u, v) = (-N, N)$ this equals $-N$, whereas the direct computation of $G_{M_1}(0, 1)$ is $-\inf_s s(s-1) = 1/4$.

The theorem gives a complete description of P_M, G_M, F_M for every positive scalar graph, rather than merely one isolated witness. It does not attempt to characterize all CMQd subsets on which equality holds.

The script `code/verify_counterexample.py` independently checks every completed-square identity and the explicit witness in exact symbolic arithmetic. The analytic proof above does not depend on computation.

A bounded novelty check was performed on 11 August 2026. The run's cheap indexes were searched by arXiv id, exact conjecture text, title, CMQd, quasidensity, and the pair G_M, F_M; the local citing corpus and targeted web/arXiv searches were also checked. No later proof, counterexample, or matching scalar computation was located. This is bounded negative evidence, not a priority claim.

Human-review recommendation. The result is elementary enough for direct line-by-line review. The critical checks are that the source uses the displayed definition of F_M with its leading minus sign, and that the codomain explicitly allows $+\infty$.

References

[1] Stephen Simons, *Faces of quasidensity*, arXiv:2407.00479 (2024), `https://arxiv.org/abs/2407.00479`.